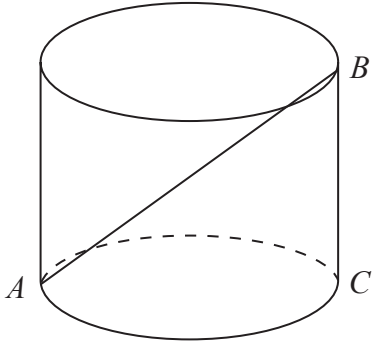
12 The diagram shows a metal rod, AB, resting inside a cylindrical tin.



The tin is on a horizontal table.
AC is a diameter of the base of the tin.
B is on the top edge of the tin.
BC is vertical.

The radius of the base of the tin is 5 cm.
The volume of the tin is 1178 cm^3

Find the angle between the rod and the base of the tin.
Give your answer correct to the nearest degree.

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(Total for Question 12 is 4 marks)